

Non-perturbative Confirmation of the Wilson–Fisher Fixed Point via Gradient Flow in 3D $O(N)$ Sigma Model

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- H. Makino, OM and H. Suzuki, PTEP **2018**, no.5, 053B02 (2018) [arXiv:1802.07897 [hep-th]] [Erratum: PTEP **2021**, no.9, 099201 (2021)]
- OM, M. Tanaka, M. Kitazawa and H. Suzuki, PoS **LATTICE2024**, 349 (2025) [arXiv:2410.19425 [hep-lat]]
- OM and H. Suzuki, arXiv:2412.03888 [hep-lat]

Memories

- Lüscher's discussions about Gradient Flow [['10-'13](#)]
 - ▶ Small flow time provides energy-momentum tensor on lattice [[Suzuki '13](#), [Makino–Suzuki '14](#)].
 - ▶ Gradient flow is similar to renormalization group flow.
- H. Makino, OM and H. Suzuki, '18 [Erratum: '21]
 - ▶ M1 @Kyushu U (2016): Polchinski' textbook, lattice chiral gauge theory [[Makino–Morikawa](#)], and this result.
 - ▶ (なんやかんやで) appeared in arXiv and PTEP, 2018.
- OM, M. Tanaka, M. Kitazawa and H. Suzuki, LATTICE2024
 - ▶ Postdoc 1st year @Osaka U (2021): Tanaka's Master Thesis (and above erratum thanks to Tanaka and Kitazawa).
 - ▶ (なんやかんやで) talked at Lattice2024.
- M. Tanaka, M. Kitazawa, OM, and H. Suzuki, in preparation/work in progress

- 1 Introduction
- 2 Review of (perturbative) gradient flow
- 3 Gradient flow and Wilsonian RG
- 4 Analytical computation of RG flow via gradient flow
 - 4D N_f -flavor gauge theory
 - 3D $O(N)$ linear sigma model
- 5 Lattice simulation of RG flow via gradient flow
- 6 Discussions
 - Lattice artifacts, finite size effect
 - Winding/topological number
- 7 Summary

Introduction

- Renormalization group (RG) is characterized by scaling relation
- (Perturbation) Even if no apparent mass-scale, introduce regularizations/renormalization due to UV divergence

★ See [H. Murayama's lecture "Note on regularization."]

- Underlying physical laws depend on energy scale, say, μ or cutoff Λ

- ▶ Change as $\Lambda_\tau = e^{-\tau} \Lambda_0$, then theory $S(\tau)$.
- ▶ Microscopic laws in $S(0)$ to macroscopic phenomena by $S(\tau)$.
- ▶ Especially, useful for strongly coupled theories (QCD).

- To obtain QFT in continuum spacetime in a *constructive* sense, $\Lambda_0 \rightarrow \infty$ and see typical length " $\xi = \xi_0(K - K_C)^{1/\nu}$."

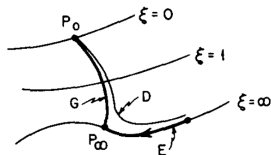


Fig. 12.6. Renormalization group trajectories near two fixed points. The curve D begins close to the critical surface. As the origin of D approaches the critical surface, the trajectory D approaches the trajectories E and G . For this example G is $S(\infty)$.

[Wilson–Kogut '74]

Non-perturbative RG idea

- Introduce cutoff function, e.g.,

$$K(p/\Lambda) = \begin{cases} 0 & \text{if } p/\Lambda \gtrsim 1, \\ 1 & \text{if } p/\Lambda \lesssim 1 \end{cases}$$

- S_Λ is a theory where all modes with $p > \Lambda$ are integrated out
- Exact functional differential equation (“Polchinski” equation)

$$-\frac{\partial}{\partial \ln \Lambda} e^{S_\Lambda} = \dots$$

- Beyond perturbation theory, **non-perturbative** construction of quantum field theory
 - ▶ Nontrivial fixed point, anomalous dimensions
- (Abstract Wilson’s) Topological idea of theory (sub)space

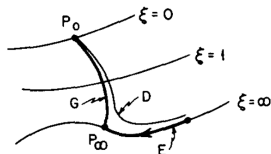


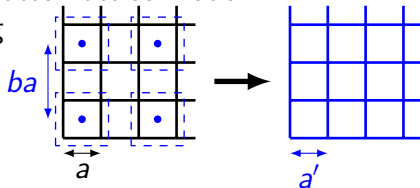
Fig. 12.6. Renormalization group trajectories near two fixed points. The curve D begins close to the critical surface. As the origin of D approaches the critical surface, the trajectory D approaches the trajectories E and G. For this example G is $S(\infty)$.

[Wilson–Kogut '74]

Lattice construction and spin blocking

- $\Lambda = 1$ and large enough τ : starting from a “lattice theory”
- Condensed matter lattice model:

Spin blocking



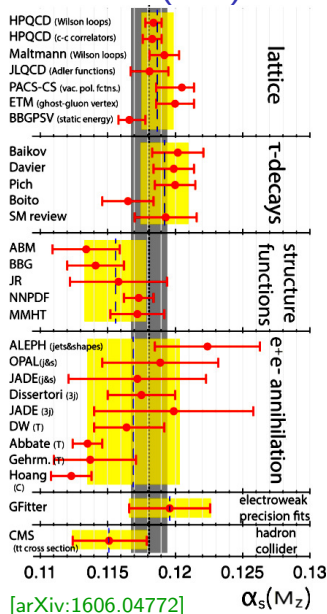
- ▶ Taking $a = 1$ and after spin blocking, we set $a' = 1$.
- ▶ Correlation length $\xi_b \sim \xi/b \rightarrow 0$ or ∞ under

$$\varphi_b(x_b) \sim \sum_{\epsilon} \varphi(x + \epsilon)$$

- Non-perturbative field theory via naive block spin
 - ▶ requires a momentum cutoff;
 - ▶ suffers from crucial problem because it is incompatible with **gauge symmetry** (Λ breaks it explicitly).

Developments along large Gradient Flow (GF)

- Lattice **gauge** theory w/ GF
 - ▶ Determination of $\alpha_s = \bar{g}^2/4\pi$
- PDG '17: $\alpha_s(M_Z) = 0.1182(12)$
- ALPHA '17: $\alpha_s(M_Z) = 0.11852(84)$
- **Precise** determination for phenomenology (see later)
 - ▶ Lattice effective action
 - ▶ Many flavor QCD ($\sim$ conformal)
- Geometry (AdS induced metric)
- ϵ expansion
- GF Exact RG (higher derivative extension)
- ...



In this talk, I will

- Provide GF as a gauge-invariant renormalization scheme
 - Give a simple picture of Wilsonian RG flow
 - Depict some examples of RG flow with nontrivial fixed points
 - ▶ Analytically for pert. QCD and large N expansion.
 - ▶ Numerically for scalar field theory.
 - Argue its critical exponents (hopefully)
 - How about gravity (asymptotic safety) and Ricci flow
 - (My) Some questions related to categorical symmetry
 - ▶ Bockstein spectral sequence [Gagliano–Grigoletto–Ohmori '25].
- $$\mathbb{Z}_N \rightarrow U(1)_Q \rightarrow U(1)_B \rightarrow B\mathbb{Z}_N \rightarrow \dots$$
- ▶ Modular fusion category between rational CFTs [Kikuchi '24].
 - ▶ ...

Gradient flow [Lüscher 2010]

- **Gradient flow** [Narayanan–Neuberger, Lüscher]: an evolution of physical quantities along a *fictitious* time
- Gradient flow equation (gauge field) is given by

$$\partial_t B_\mu(t, x) = - g_0^2 \frac{\delta S_{\text{YM}}}{\delta A_\mu} \Big|_{A_\mu \rightarrow B_\mu} = D_\nu G_{\nu\mu}(t, x),$$

$$G_{\mu\nu}(t, x) = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu]$$

where $G_{\mu\nu} = \partial_\mu B_\nu - \partial_\nu B_\mu + [B_\mu, B_\nu]$ and the boundary condition

$$B_\mu(t = 0, x) = A_\mu(x)$$

- t : mass dimension -2
- “Gradient” $\rightarrow \partial_t B_\mu = -g_0^2 \delta S_{\text{YM}} / \delta B_\mu$
- Diffusion equation with the diffusion length

$$x \sim \sqrt{8t}$$

Smoothing effect for the configuration as $t \rightarrow \text{large}$

Backup: Perturbative expansion of GF [1/2]

- Gradient flow with the “gauge fixing”

$$\partial_t B_\mu(t, x) = D_\nu G_{\nu\mu}(t, x) + \alpha_0 D_\mu \partial_\nu B_\nu(t, x)$$

- Formal solution of this equation

$$B_\mu(t, x) = \int d^D y \left[K_t(x - y)_{\mu\nu} A_\nu(y) + \int_0^t ds K_{t-s}(x - y)_{\mu\nu} R_\nu(s, y) \right]$$

where the heat kernel

$$\begin{aligned} K_t(x)_{\mu\nu} &= \int \frac{d^d p}{(2\pi)^d} \frac{e^{ipx}}{p^2} \left[(\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-tp^2} + p_\mu p_\nu e^{-\alpha_0 t p^2} \right] \\ &= \delta_{\mu\nu} \int \frac{d^d p}{(2\pi)^d} e^{ipx} e^{-tp^2} \quad (\alpha_0 = 1) \end{aligned}$$

- and the non-linear term R

$$R_\mu = 2[B_\nu, \partial_\nu B_\mu] - [B_\nu, \partial_\mu B_\nu] + (\alpha_0 - 1)[B_\mu, \partial_\nu B_\nu] + [B_\nu, [B_\nu, B_\mu]]$$

Backup: Perturbative expansion of GF [2/2]

- Correlation function of the flowed gauge field

$$\begin{aligned} & \langle B_{\mu_1}(t_1, x_1) \dots B_{\mu_N}(t_N, x_N) \rangle \\ &= \frac{1}{\mathcal{Z}} \int \mathcal{D}A_\mu B_{\mu_1}(t_1, x_1) \dots B_{\mu_N}(t_N, x_N) e^{-S_{\text{YM}} - S_{\text{gf}} - S_{c\bar{c}}} \end{aligned}$$

- Free propagator of the flowed field

$$\begin{aligned} & \langle B_\mu^a(t, x) B_\nu^b(s, y) \rangle_0 \\ &= \delta^{ab} g_0^2 \int \frac{d^d p}{(2\pi)^d} \frac{e^{ip(x-y)}}{(p^2)^2} \left[(\delta_{\mu\nu} p^2 - p_\mu p_\nu) e^{-(t+s)p^2} + p_\mu p_\nu e^{-\alpha_0(t+s)p^2} \right] \\ &= \delta^{ab} \delta_{\mu\nu} g_0^2 \int \frac{d^d p}{(2\pi)^d} \frac{e^{ip(x-y)}}{p^2} e^{-(t+s)p^2} \quad (\alpha_0 = 1) \end{aligned}$$

- Gaussian damping factor e^{-tp^2}

$\Rightarrow$ a simple renormalization property of the gradient flow

Flowed composite operator [Lüscher–Weisz 2011]

- Correlation function of the flowed gauge field ($t_i > 0$)

$$\langle B_{\mu_1}(t_1, x_1) \dots B_{\mu_N}(t_N, x_N) \rangle$$

⇒ **UV finite** without wave function renormalization

▶ Momentum integral $\int d^D p \rightarrow \int d^D p \exp(-tp^2)$

- Independence of regularization

→ Probes of universal properties of theories

- Compatible with **gauge invariance**

- Many interesting applications to lattice gauge theory

- ▶ Topological charge
- ▶ Chiral condensation
- ▶ Energy-momentum tensor
- ▶ Non-perturbative gauge coupling
- ▶ (cf. Normalizing flow $\overset{?}{\sim}$ trivializing gradient flows)

Flow of matter field [Lüscher 2013]

- A possible choice of flow equation

$$\begin{aligned}\partial_t \chi(t, x) &= [\Delta - \alpha_0 \partial_\mu B_\mu(t, x)] \chi(t, x), & \chi(t=0, x) &= \psi(x), \\ \partial_t \bar{\chi}(t, x) &= \bar{\chi}(t, x) \left[\overleftarrow{\Delta} + \alpha_0 \partial_\mu B_\mu(t, x) \right], & \bar{\chi}(t=0, x) &= \bar{\psi}(x),\end{aligned}$$

where

$$\begin{aligned}\Delta &= D_\mu D_\mu & D_\mu &= \partial_\mu + B_\mu \\ \overleftarrow{\Delta} &= \overleftarrow{D}_\mu \overleftarrow{D}_\mu & \overleftarrow{D}_\mu &= \overleftarrow{\partial}_\mu - B_\mu\end{aligned}$$

- Now, we need the wave function renormalization:

$$\chi_R(t, x) = Z_\chi^{1/2} \chi(t, x) \qquad \bar{\chi}_R(t, x) = Z_\chi^{1/2} \bar{\chi}(t, x)$$

- Any composite operators of χ_R , $\bar{\chi}_R$ are **UV finite**

Gradient flow and block spin transformation?

- Apparently, gradient flow acts as “coarse-graining” [Lüscher ...]
- E.g., Yang–Mills gradient flow equation (not pert.)

$$\partial_t B_\mu(t, x) = - \left. \frac{\delta S_{\text{YM}}}{\delta A_\mu} \right|_{A_\mu \rightarrow B_\mu} = D_\nu G_{\nu\mu}(t, x)$$

- Diffusion eq. $\partial_t \phi(t, x) = \partial_\mu^2 \phi(t, x)$ w/ diffusion length $x \sim \sqrt{8t}$
- Blocked field is given as
- Solution of $\partial_t \phi(t, x) = \partial_\mu^2 \phi(t, x)$

$$\varphi_b(x_b) \sim \sum_{\epsilon} \varphi(x + \epsilon)$$

▶ $b \sim \sqrt{8t}$?

$$\phi(t, x) \sim \int_y e^{-(x-y)^2/4t} \varphi(y)$$

- Can we define *gauge-invariant* Wilson's RG by gradient flow?

Wilsonian RG flow [Makino–OM–Suzuki '18]

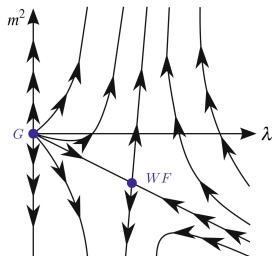
- Scaling relation under Wilsonian RG (running coupling $g_i(\xi)$)

$$\begin{aligned} & \langle \mathcal{O}_1(e^{2\xi} t_1, e^\xi x_1) \dots \mathcal{O}_N(e^{2\xi} t_N, e^\xi x_N) \rangle_{\{g_i\}} \\ &= Z(\xi) \langle \mathcal{O}_1(t_1, x_1) \dots \mathcal{O}_N(t_N, x_N) \rangle_{\{g_i(\xi)\}} \end{aligned}$$

- Define new scheme as $g_{\text{GF}}^2(\mu) = \langle \mathcal{O}(t) \rangle_{\mu=1/\sqrt{8t}} \stackrel{\text{pert.}}{\sim} g^2 + O(g^4)$
- Assumption as one-to-one map $g_i(\xi) \mapsto \langle \mathcal{O}_i(t) \rangle = \mathcal{R}_i[\{g_j(\xi)\}]$

Gradient flow $\langle \mathcal{O}_j(e^{2\xi} t) \rangle_{\{g_i\}} = \langle \mathcal{O}_j(t) \rangle_{\{g_i(\xi)\}}$ Running

- Gradient flow is gauge covariant!
 - ▶ A gauge-invariant construction of non-perturbative RG flow



Cf. Numerical determination of α_S in QCD

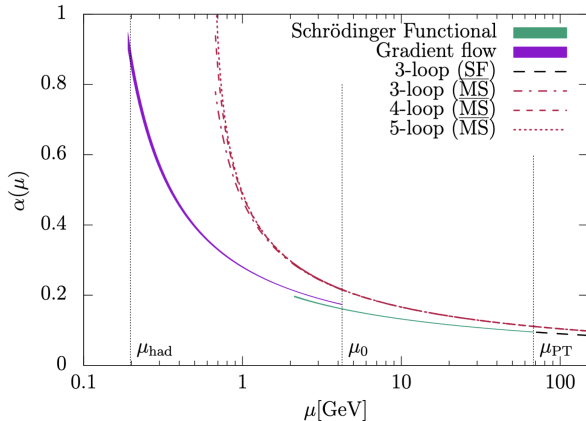


FIG. 2. Running couplings of $N_f = 3$ QCD from integrating the nonperturbative β -functions in the SF and GF schemes [13, 14]. They are matched nonperturbatively by defining $\bar{g}_{\text{SF}}^2(\mu_0) = 2.012$ and computing $\bar{g}_{\text{GF}}^2(\mu_0/2) = 2.6723(64)$.

[ALPHA '17]

Analytical computation of RG flow via GF

- ① Banks–Zaks fixed point in (2-loop approx) N_f -flavor QCD
- ② Wilson–Fisher fixed point in large N 3D $O(N)$ linear sigma model

- ① 4D gauge theory with N_f -flavor Dirac fermions (mass m)
 - Banks–Zaks IR fixed point [’82]
 - Running coupling and mass parameter in the $\overline{\text{MS}}$ scheme

$$\left[b_0 \bar{g}(\mu)^2\right]^{-b_1/(2b_0^2)} \exp\left[-\frac{1}{2b_0 \bar{g}(\mu)^2}\right] = \frac{\Lambda}{\mu},$$
$$\bar{m}(\mu) = M \left[2b_0 \bar{g}(\mu)^2\right]^{d_0/(2b_0)},$$

where

$$b_0 = \frac{1}{(4\pi)^2} \left[\frac{11}{3} C_2(G) - \frac{4}{3} T(R) N_f \right],$$
$$b_1 = \frac{1}{(4\pi)^2} \left\{ \frac{34}{3} C_2(G)^2 - \left[4C_2(R) + \frac{20}{3} C_2(G) \right] T(R) N_f \right\}, \quad d_0 = \frac{1}{(4\pi)^2} 6C_2(R)$$

(Λ and M are RG invariant mass scales)

4D N_f -flavor gauge theory

- Dimensionless operators

$$\mathcal{O}_1(t, x) \equiv \frac{8(4\pi)^2 t^2}{3 \dim(G)} \frac{1}{4} G_{\mu\nu}^a(t, x) G_{\mu\nu}^a(t, x)$$

$$\mathcal{O}_2(t, x) \equiv \frac{\bar{\chi}(t, x) \chi(t, x)}{t^{1/2} \left\langle \bar{\chi}(t, x) \overleftrightarrow{D} \chi(t, x) \right\rangle}$$

- Two-loop approximation in $\overline{\text{MS}}$ scheme [Harlander–Neumann]

$$\langle \mathcal{O}_1(t) \rangle = \bar{g}(1/\sqrt{8t})^2 \left[1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} K_1(t) + \frac{\bar{g}(1/\sqrt{8t})^4}{(4\pi)^4} K_2 \right],$$

where (leading mass correction only)

$$K_1(t) = \left(\frac{11}{3} \gamma_E + \frac{52}{9} - 3 \ln 3 \right) C_2(G) \left[-\frac{4}{3} \gamma_E - \frac{8}{9} + \frac{8}{3} \ln 2 + 16 \bar{m}(1/\sqrt{8t})^2 t \right] T(R) N_f,$$

$$K_2 = 8(4\pi)^2 \{ -0.0136423(7) C_2(G)^2 \\ + [0.006440134(5) C_2(R) - 0.0086884(2) C_2(G)] T(R) N_f + 0.000936117 T(R)^2 N_f^2 \}$$

- One-loop approximation [Makino–Suzuki]

$$\langle \mathcal{O}_2(t) \rangle = \bar{m}(1/\sqrt{8t}) t^{1/2} \left[1 + \frac{\bar{g}(1/\sqrt{8t})^2}{(4\pi)^2} K'_1 \right], \quad K'_1 = [3\gamma_E + 4 + 2 \ln 2 - \ln(432)] C_2(R)$$

4D N_f -flavor gauge theory

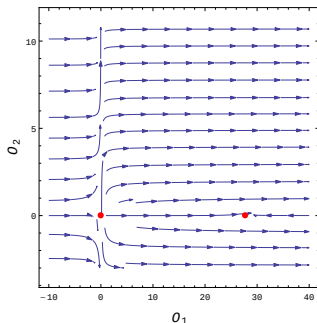
- Running parameters

$$\begin{aligned} t \frac{d}{dt} \langle \mathcal{O}_1(t) \rangle &= b_0 \langle \mathcal{O}_1(t) \rangle^2 + b_1 \langle \mathcal{O}_1(t) \rangle^3 \\ &\quad + \frac{1}{(4\pi)^2} 16 T(R) N_f \langle \mathcal{O}_1(t) \rangle^2 \langle \mathcal{O}_2(t) \rangle^2 \end{aligned}$$

$$\begin{aligned} t \frac{d}{dt} \langle \mathcal{O}_2(t) \rangle &= \frac{1}{2} [1 + d_0 \langle \mathcal{O}_1(t) \rangle] \langle \mathcal{O}_2(t) \rangle \end{aligned}$$

- In the IR limit $t \rightarrow \infty$, $\langle \mathcal{O}_2(t) \rangle \rightarrow \infty$: relevant coupling
- Bank-Zaks fixed point:

$$(\langle \mathcal{O}_1(t) \rangle, \langle \mathcal{O}_2(t) \rangle) = (-b_0/b_1, 0)$$



3D $O(N)$ linear σ -model at large N

- Action of the 3D $O(N)$ linear σ -model

$$S = \int d^3x \left\{ \frac{1}{2} \partial_\mu \phi_i(x) \partial_\mu \phi_i(x) + \frac{1}{2} m_0^2 \phi_i(x) \phi_i(x) + \frac{\lambda_0}{8N} [\phi_i(x) \phi_i(x)]^2 \right\}$$

→ **Wilson–Fisher IR fixed point** ['72]

- For simplicity, we will work out the large- N approximation
 - ▶ Gradient flow in this system was studied in [Aoki et al. '17]
- “Physical” mass M given by the gap equation

$$M^2 + \frac{\lambda_0}{8\pi} M = m_0^2 + \frac{1}{4\pi^2} \lambda_0 \Lambda$$

(Λ : momentum cutoff)

- Flow of $\phi_i(x)$

$$\partial_t \varphi_i(t, x) = \partial_\mu \partial_\mu \varphi_i(t, x), \quad \varphi_i(t = 0, x) = \phi_i(x)$$

3D $O(N)$ linear σ -model at large N

- Dimensionless operators

$$\mathcal{O}_1(t, x) \equiv -\frac{N [\varphi_i(t, x) \varphi_i(t, x)]^2}{\langle \varphi_j(t, x) \varphi_j(t, x) \rangle^2} + (N + 2),$$

$$\mathcal{O}_2(t, x) \equiv \frac{4t \partial_\mu \varphi_i(t, x) \partial_\mu \varphi_i(t, x)}{(2\pi)^{1/2} \langle \varphi_j(t, x) \varphi_j(t, x) \rangle} - \frac{1}{(2\pi)^{1/2}}$$

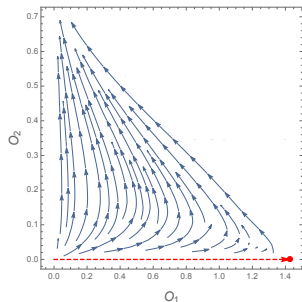
- At the leading order of the large- N expansion,

$$\begin{aligned} \langle \mathcal{O}_1(t) \rangle &= \lambda_0 t^{1/2} \left[\int \frac{d^3 p}{(2\pi)^3} \frac{e^{-2p^2}}{p^2 + M^2 t} \right]^{-2} \\ &\quad \times \prod_{i=1}^4 \left(\int \frac{d^3 p_i}{(2\pi)^3} \frac{e^{-p_i^2}}{p_i^2 + M^2 t} \right) (2\pi)^3 \delta(p_1 + p_2 + p_3 + p_4) \\ &\quad \times \left[1 + \frac{\lambda_0 t^{1/2}}{8\pi} \frac{1}{\sqrt{(p_1 + p_2)^2}} \arctan \left(\frac{1}{2} \sqrt{\frac{(p_1 + p_2)^2}{M^2 t}} \right) \right]^{-1} \\ \langle \mathcal{O}_2(t) \rangle &= \frac{1}{8\pi^2} \left(\int \frac{d^3 p}{(2\pi)^3} \frac{e^{-2p^2}}{p^2 + M^2 t} \right) - \left(\frac{8}{\pi} \right)^{1/2} M^2 t - \frac{1}{(2\pi)^{1/2}} \end{aligned}$$

3D $O(N)$ linear σ -model at large N

- Asymptotic behaviors:

$$\begin{aligned}\mathcal{O}_1(t, x) &\equiv -\frac{N [\varphi_i(t, x)^2]^2}{\langle \varphi_j(t, x)^2 \rangle^2} + (N + 2) \\ &\stackrel{t \rightarrow 0}{\sim} \lambda_0 t^{1/2} \\ &\stackrel{t \rightarrow \infty}{\rightarrow} \frac{1}{(4\pi)^{3/2}} \frac{\lambda_0}{M} \left(1 + \frac{1}{16\pi} \frac{\lambda_0}{M} \right) \frac{1}{M^3 t^{3/2}}, \\ \mathcal{O}_2(t, x) &\equiv \frac{4t \partial_\mu \varphi_i(t, x)^2}{(2\pi)^{1/2} \langle \varphi_j(t, x)^2 \rangle} - \frac{1}{(2\pi)^{1/2}} \\ &\stackrel{t \rightarrow 0}{\sim} M t^{1/2} \\ &\stackrel{t \rightarrow \infty}{\rightarrow} \left(\frac{2}{\pi} \right)^{1/2} - \frac{3}{(8\pi)^{1/2}} \frac{1}{M^2 t}\end{aligned}$$



[Makino–OM–Suzuki '18]

- IR fixed point at $M/\lambda_0 \rightarrow 0$
 - cf. [Aoki–Balog–Onogi–Weisz '16]

3D $O(N)$ linear σ -model at large N

- Gaussian fixed point $(0, 0)$ at small λ_0/M
 - ▶ In the $\lambda_0 \rightarrow 0$ limit

$$\langle \mathcal{O}_1(t) \rangle_G \propto \lambda_0 t^{1/2}, \quad \langle \mathcal{O}_2(t) \rangle_G \propto M t^{1/2},$$

and therefore

$$\left(t \frac{d}{dt} \right) \langle \mathcal{O}_i(t) \rangle_G = \frac{1}{2} \langle \mathcal{O}_i(t) \rangle_G + O(\langle \mathcal{O}_i(t) \rangle_G^2).$$

- Wilson–Fisher $(K_*, 0)$ at large λ_0/M
 - ▶ Irrelevant behavior in the $M \rightarrow 0$ limit

$$\langle \mathcal{O}_1(t) \rangle_{\text{WF}} - K_* \propto \lambda_0^{-1/2} t^{-1/2},$$

and the relevant one in the $\lambda_0 \rightarrow \infty$ limit

$$\langle \mathcal{O}_1(t) \rangle_{\text{WF}} - K_* \propto M t^{1/2}, \quad \langle \mathcal{O}_2(t) \rangle_{\text{WF}} \propto M t^{1/2}.$$

- ▶ After diagonalization,

$$\left(t \frac{d}{dt} \right) \langle \mathcal{O}_{1,2}(t) \rangle_{\text{WF}} = \mp \frac{1}{2} \langle \mathcal{O}_{1,2}(t) \rangle_{\text{WF}} + O(\langle \mathcal{O}_{1,2}(t) \rangle_{\text{WF}}^2).$$

Lattice setup with finite N

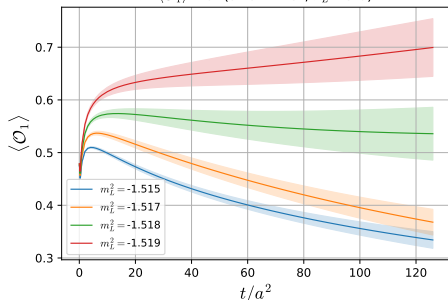
- Parameter set

- ▶ Lattice $64^3, 128^3$
- ▶ $N = 1$ (partially 2, 3, 5)
- ▶ Tuning $\lambda_0 a, m_0 a$ for small M

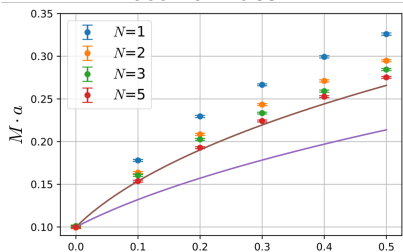
- Numerical simulation

- ▶ Overrelaxed heatbath
- ▶ 4th order Runge–Kutta method for gradient flow

$t - \langle \mathcal{O}_1 \rangle$ Plot (size = $128^3, \lambda_L = 5.0$)

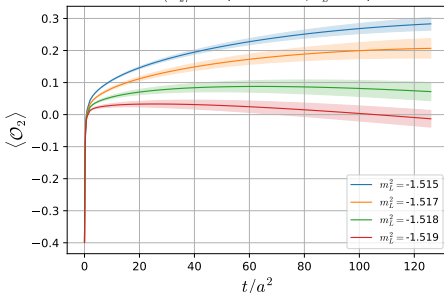


Effective mass M



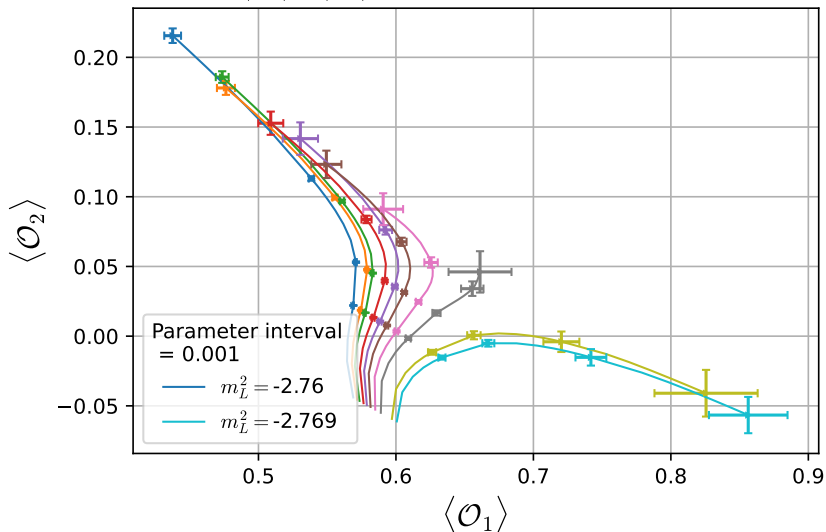
Two curves given by gap eq. at $N \rightarrow \infty$
with $\Lambda = \pi/2a$ or π/a

$t - \langle \mathcal{O}_2 \rangle$ Plot (size = $128^3, \lambda_L = 5.0$)



Lattice simulation of RG flow near criticality

$\langle \mathcal{O}_1 \rangle - \langle \mathcal{O}_2 \rangle$ Plot (size = 128^3 , $\lambda_L = 10.0$)

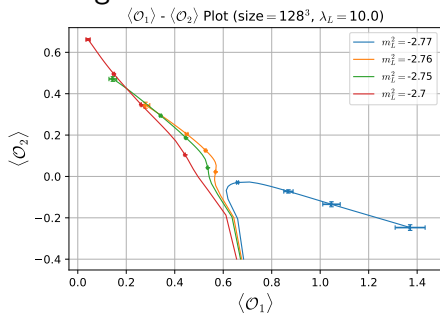


- Discussion: Finite size effect, Gaussian fixed point (following slides)

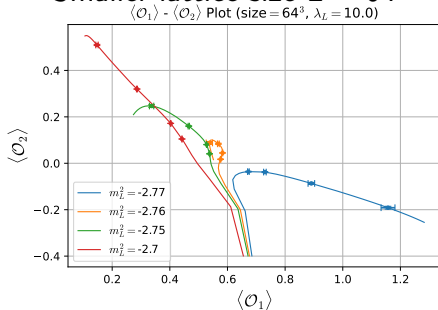
Finite lattice size effect

- Some trajectories in RG flow are intersecting near IR.
- This is because of a finite size effect as follows:

Larger lattice size $L = 128$

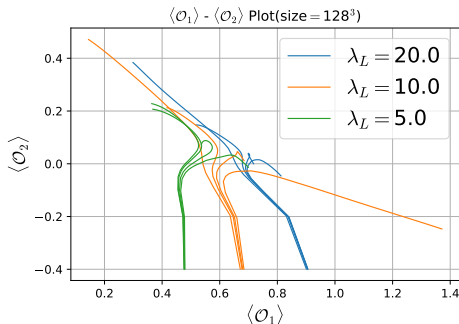


Smaller lattice size $L = 64$



From different “lattice models” to WF fixed point

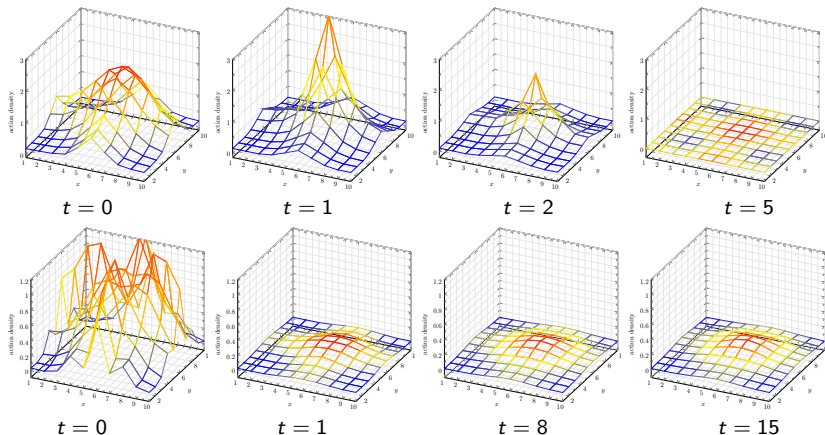
- Starts from different points at UV (not Gaussian)
 - ▶ Not continuum limit (i.e., Gaussian) but just a lattice model.
 - ▶ Critical behaviors for each models should be same near WF.



- Small λ : close to Gaussian and far from WF
 - ▶ Much bigger box size and reducing errors at large t
- Large λ : far from Gaussian but hopefully close to WF
 - ▶ More computational costs when generating configurations

More issue: Winding/topological number

- GF is a kind of smearing whereas configuration can be noisy
- For large t , is winding in lower dim or instanton stable?
 - ▶ Continuum flow stabilizes it; discretization spoil it usually.
 - ▶ “Over-improved” discretization [Tanizaki–Tomiya–Watanabe '24, OM–Suzuki '24] may stabilize topological number.



Summary

- Depicted RG flow near criticality by using operators evolved by gradient flow
 - ▶ Banks–Zaks/Wilson–Fisher fixed point at IR ($t \rightarrow \infty$)
- Numerical simulation of 3D $O(N)$ linear sigma model
- More understanding on Gradient flow and RG in **gauge theory!**
 - ▶ Manifestly gauge-invariant construction of QFT
 - ▶ See Gradient-flow Exact (Functional) RG
[Abe–Fukuma Carosso–Hasenfratz–Neil, Sonoda–Suzuki, etc.]
 - ▶ Application to lattice simulation of gauge theory?
- Critical exponents and finite size scaling (wip)
- ((De)Stabilization of winding/topological charge
[Kanamori–OM–Nagai–Tanizaki–Tomiya–Watanabe for YM,
Fukui–OM–Suzuki for condensed matter])
- Gravity (asymptotic safety and Ricci flow)? Category?
- Wilsonian philosophy: every quantum system *sub specie aeternitatis* is clarified as a continuum theory near criticality